

Define Gait :-

It is a translatory progression of body as a whole produced by coordinated rotatory movements of body segments (various action or motions of the body parts).

Define Gait Cycle :-

Time period or sequence of events during locomotion in which one foot comes in contact with the ground to the time when the same foot comes in contact with the ground again.

Major tasks of gait :-

→ 5 main tasks for walking gait.

1. Maintenance of support of the head, arms, trunk, that is, preventing collapse of the lower limb.
2. Maintenance of upright posture and balance of the body.
3. Control of the foot trajectory to achieve safe ground clearance and a gentle heel or toe landing.
4. Generation of mechanical energy to maintain the present forward velocity or to increase the forward velocity.
5. Absorption of mechanical energy for shock absorption and stability or to decrease the forward velocity of the body.

→ Maintaining balance and stability is clearly imp during ambulation, and there is an ↑ing body of literature on what might be called sub-tasks of gait that are potentially destabilizing.

Phases of the Gait Cycle :-

→ A gait cycle spans 2 successive events of the same limb, usually initial contact of the lower extremity with the supporting surface.

→ During gait cycle, each extremity passes 2 major phases

- ① Stance phase
- ② Swing phase

→ Stance phase :

Some part of the foot is in contact with the ground, it makes 60% of gait cycle.

Swing phase :

Foot is not in contact with the ground, which makes up the remaining 40%.

→ There are 2 period of double support,

- One limb makes initial contact
- Another one leave the foot at toe off.

→ Normal walking Speed :

(20%) Double Support : occupies about 11% of gait cycle, it makes total of 22% for a full cycle.

(40%) Single Support : Body supported by one limb for 80% of the cycle.

→ Value of 10% for each double-support phase is usually assigned to each of the 2 double-support periods.

→ Gait cycle is divided into events and phases.

Events in Stance Phase :-

① Initial contact :- (Heel Strike phase)
The instant the foot of the leading extremity strikes the ground.

Normal gait :-

Heel is the point of contact.
Event referred to heel contact or heel strike.

Abnormal gait :-

It is possible for the whole foot or the toe, rather than the heel, to make initial contact with the ground.

② Flat foot (or) Foot flat :-

Occurs after initial contact at approximately 7% of the gait cycle.

1st instant during stance when the foot is flat on the ground.

③ Midstance :-

Body wt is directly over the supporting lower extremity, usually about 30% of the gait cycle.

④ Heel-off :-

The heel of the reference extremity leaves the ground, usually about 40% of the gait cycle.

⑤ Toe off :-

Toe of the foot leave the ground, 60% of gait cycle.

% Gait Cycle	10%	10%	20%	30%	40%	50%
Event	initial Contact 0%	Foot flat 7%		midstance 30%	Heel off 40%	
Phase	Heel strike	Midstance			Push off	

Subphases of Stance Phase :-

1) Heel strike phase -

- begins with initial contact and ends with foot flat, occupies small % of gait cycle.
 - Terminology Center includes this phase in one called weight acceptance, or loading response.
- Begins at initial contact and ends when the contralateral extremity lifts off the ground at the end of the double support phase, occupies 11% of gait cycle.

2) Midstance phase -

- Begins with foot flat at 7% of gait cycle & ends with heel off at 40% of gait cycle.

3) Push off phase -

- Begins with heel off, 40% of gait cycle & ends with toe off at 60% of gait cycle.

→ Terminology originating in Karano Los Amigos National Rehabilitation Center - refers to the last 10% of stance phase, beginning with initial contact of the contralateral foot at 50% of gait cycle and ending with foot flat at 60% of gait cycle.

Swing Phase :-

1) Early Swing Phase -

→ Begins once the toe leaves the ground & continues until midswing. Occurs at 60-75%.
This phase is referred to as initial swing or the acceleration phase.

2) Midswing -

→ When the extremity passes directly beneath the body, or from the end of acceleration to the beginning of deceleration.

→ Occurs 75 to 85%.

3) Late swing -

→ When the limb is decelerating in preparation for heel strike.

→ Also known as terminal swing or the deceleration phase

→ Occurs 85 to 100%.

Gait Terminology :-

Time and Distance Terms -

→ Time & Distance are 2 basic parameters of motion.

→ Time or Temporal variables :-

includes ; Stance time, Single-limb and double limb support time, Swing time, stride & step time, Cadence and speed.

→ Distance Variables includes ;

Stride length, Step length, width, degree of toe out.

Stance time -

→ Amount of time that elapses during the stance phase of one extremity in a gait cycle.

Single - Support time -

→ Amount of time that elapses during the period, when 1 extremity is on the supporting surface in gait cycle.

Double - Support time -

→ Amount of time spent with both feet on the ground during 1 gait cycle.

→ % of time in double support ↑ in elderly persons.

→ % of time in double support ↓ as speed of walking ↑.

Stride length -

→ Linear distance b/w 2 successive events by same lower extremity.

→ "Linear distance from the point of 1 heel strike of the lower extremity to the point of next heel strike of the same extremity."

→ The length of one stride includes all the events of gait cycle.

→ Stride length ↓ in elderly persons, ↑ as the speed of gait ↑.

Stride duration -

→ Amount of time is taken to accomplish 1 stride.

Step length -

→ Linear distance b/w 2 successive points of contact of opp extremity.

→ It is measured from the heel strike of one extremity to the heel strike of another extremity.

Step duration -

→ Amount of time spent during a single step.

→ Measurement usually, is expressed as seconds per step.

→ When the leg is weakness or pain, step duration ↓ on affected side and ↑ on unaffected side.

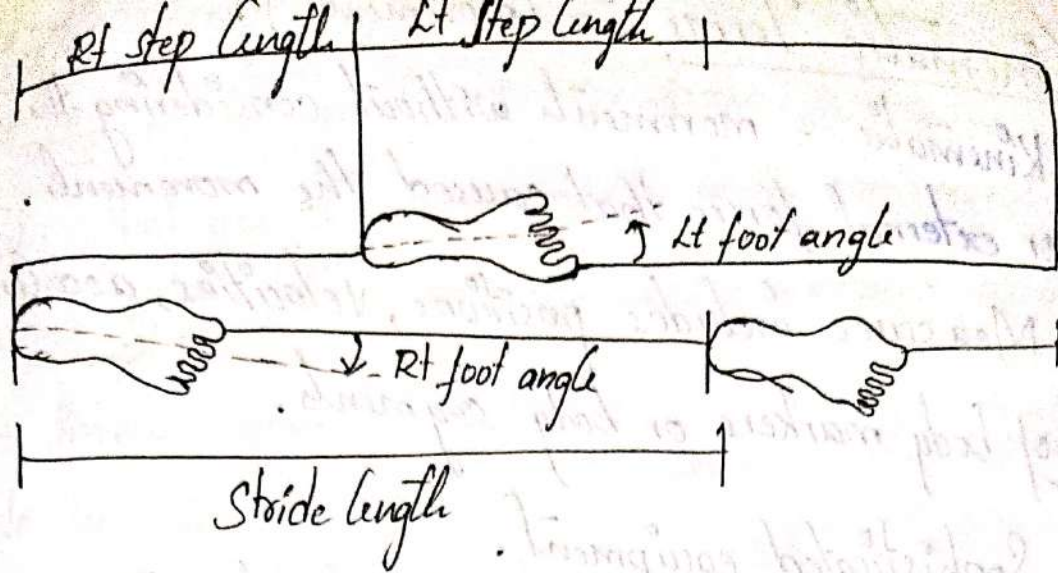
Cadence -

- No. of Steps taken by a person per unit of time.
- Measured as the no. of steps per second or per minute.
- Shorter step length will result in an $\uparrow$ cadence at any given velocity.
- When a person walks with a cadence b/w 80 to 120 steps per minute, Cadence and stride length had a linear relationship.
- As a person walks with $\uparrow$ cadence, the duration of the double support period $\downarrow$ s.
- Cadence of walking approaches 180 steps per minute the period of double support disappears & running commences.
- Adult men - 110 steps per minute (cadence).
Women - 116 steps per minute (cadence).

Walking Velocity -

- It is a rate of linear forward motion of the body, which can be measured in meters or centimeters per second, meters per minute, or miles per hour.

$$\text{Walking Velocity (meters/second)} = \frac{\text{distance walked (m)}}{\text{time (sec)}}$$



Step width -

- Also known as width of the walking base.
- Measuring the linear distance b/w the mid point of the heel of one foot and the same point on the other foot.
- ↑ when there is an need demand for side-to-side stability. (Elderly persons & small children).
- In toddlers and ~~old~~ Young children, COG is higher than in adults, wide BOS is necessary for stability.
- Normal population, width of BOS is 3.5 inches & varies within a range of 1 to 5 inches.

Degree of Out-toe -

- Represents the angle of foot placement.
- Measuring the angle formed by each foot's line of progression & a line intersecting the center of the heel and the 2nd toe.
- Angle for men - 7° (foot at free speed walking).
- Degree of toe-out ↓ as the speed of walking ↑ in normal men.

Kinematic Terms :- (movements).

→ Kinematic - movements without considering the ⁱⁿ or external forces that caused the movements.

→ Measures includes positions, velocities, acceleration of body markers or body segments.

→ Sophisticated equipment -

First, Stroboscopic photography, then cinematography and electrogoniometers.

→ Observational gait analysis -

Employs a less sophisticated and less objective method in which an observer makes a judgement as to whether a particular ^{of} angle or motion arises from a norm.

→ Trajectories -

Or the paths in space of particular parts, are of particular interest.

→ Joints angles -

They are the primary importance in recording gait.

→ Anatomical position is called the "Zero" position of the ft.

Kinetic Terms :- (forces)

→ Kinetics is concerned with the forces acting on the body that are the cause of the body movement.

→ A link segment model, is most often used with an inverse dynamic approach.

→ We look at the body as the series of segments (bottom to top) (foot, lower leg, thigh -- etc).

→ Walking person, there are ground reaction forces, derived from a force platform, the wt of the body part (gravity), inertia of the body part.

= Inertia - (unit $\text{kg} \times \text{m}^2$) is a measure of an object's resistance to change in its rotation state.

Ground Reaction Forces (GRF) :-

→ Force being applied to the body foot by the ground when a person take a step.

→ These forces are equal in size (magnitude) but opposite in direction.

→ GRFs are used in vertical, anteroposterior, medio lateral axes.

→ If the force components in 2 or 3 planes and it termed as Ground Reaction Force Vector (GRFV)

→ Single component, such as the vertical GRF, GRF is used.

→ Single Vector (force with a direction), GRFV is used.

→ Center of pressure of the foot on the supporting surface is the point where the resultant of all the floor-foot forces act.

→ GRFs, GRFVs and CoPs are the characteristic pattern of normal walking.

Moment of force :-

→ The product of the force and the perpendicular distance from its action line to the pt of center.

Internal moments :-

→ They are the moments generated by the muscles, joint capsules and ligaments to counteract the external forces acting on the body.

→ External forces such as Ground reaction forces produce external moments about the joints.

Energy :-

→ It is a capacity to do work.

→ Both work & energy are expressed in the same units, joules (J).

→ Muscles use metabolic energy to perform mechanical work by converting metabolic energy into mechanical energy.

→ Metabolic cost incurred during locomotion may be measured by assessing the body's Oxygen Consumption ($\dot{V}O_2$) per unit of distance.

→ long distance is travelled only small amount of oxygen is consumed, metabolic cost of particular gait is low.

→ 32% of max oxygen consumption is needed by 20-30 yrs. old to walk at comfortable speed.

→ 48% for 75 yrs. old. And even higher if a person is suffering from chronic condition like stroke.

→ Metabolic equivalents (METs) are also used to express energy cost of the activity as multiples of the resting metabolic rate.

→ Oxygen Consumption - for a person walking at 4 to 5 km/hour avg 100 mL/kg body wt per minute, typically 2.5 to 4 mL.

→ Speed of walking increases, energy cost per unit of distance walked $\uparrow$.

→ Speed of walking $\downarrow$, energy cost $\uparrow$.

Power Generation :-

→ It is accomplished when muscles shorten (concentric contraction).

→ They do positive work and add to the total energy of the body.

→ Power, expressed in watts, is the work or energy

Value (in joules) -

→ Power of muscle groups performing gait is calculated

→ Ted through an inverse dynamic approach.

→ The most imp phases of power generation & absorption during walking have been designated by joint and plane.

Power Absorption :-

→ It is accomplished when muscles perform a lengthening (eccentric contraction).

→ They do ~~pos~~ Negative work.

Electromyography :- (EMG)

→ It is a technique for evaluating and recording the activation signal of muscles.

→ EMG is performed using an instrument called an Electromyograph, to produce a record called an Electromyogram.

→ In it electrical activity generated by a muscle is recorded.

Mechanical Energy of Walking :-

- Walking at a constant speed is accomplished by the bursts of positive (concentric) and Negative (eccentric) muscle work.
- At initial contact of each foot, at 0% ± 50% of the gait cycle, body has the lowest potential energy & it moves into midstance, potential energy rises and exchange can be for kinetic energy.

Muscle Activity :-

- The muscle work of gait can be simplified by anticipating the muscle groups that must be functioning for specific purposes.
- 2 Major features of Muscle work.
 - * The need to provide a support moment through stance.
 - * The need to generate energy to move.
- Support moment is made up of hip extensors, knee extensors and Ankle plantar/flexors. (calf)
- Hip Extensors produce extensor moment early in stance. (hamstrings)

Initial contact to Midstance.

Sagittal Plane Analyses

- Knee extensors produces a knee extensor moment.
- Flexor moments are produced at both the hip & knee before hip & knee bend in late stance for swing phase. (quadriceps)

* Note :-

Muscle activity results in effects as diverse as external locomotion of the body and the internal movement of substance in the cardiovascular and gastrointestinal systems.

Joint	Motion	GRF	Internal moment	Power	Major Muscle Activity
Hip	Extends from $+20^{\circ}$ to $+30^{\circ}$ to 0°	Ant to post	Extensor to Neutral.	General (Hip extensor)	Hamstrings Gluteus maximus
Knee	Flexes from 0° to $+15^{\circ}$ then extends to $+15^{\circ}$ to $+5^{\circ}$	Ant to post to anterior	Flexor, then extensor then flexor	General (knee flexors) Absorpt (knee extensors) General (knee extensors) Absorpt (knee flexors)	Hamstrings Quadriceps Gastrocnemius
Ankle	Plantar flexion from 0° to $+5^{\circ}$ then dorsiflexion to $+5^{\circ}$	Post to anterior	Dorsiflexor, then plantar flexor	Absorpt (dorsiflexors) General (plantar flexors)	Fibularis ant Soleus Gastrocnemius

Frontal Plane Analysis

Joint	Motion	GRF	Moment	Power	Major Muscle Activity
Hip	Add - from neutral to +5° then abd to 3°	Medial	Abductor	Absorpt ⁿ (hip abd) Generat ⁿ (hip abd)	Gluteus medius
Knee	Small variat ⁿ around Neutral	Medial	Abductors	Small generat ⁿ (knee abd)	Tensor fasciae latae
Ankle	Inconsistent	—	—	—	—

Transverse Plane Analysis

Joint	Motion	Moment	Power	Major Muscle Activity
Hip	Externally rotates a few degrees	External rotator	Absorpt ⁿ (intermed. rotators)	Tensor fasciae latae Gluteus medius/min
Knee	From several degrees of external rotation rotates internally to neutral	Very small external rotators	Small absorpt ⁿ (external rotators)	—

Mid stance to toe off

Sagittal Plane Analysis

It	Motion	GRF	Moment	Power	Major Muscle Activity
Hip	Extends from 0° to -20° Extends, then begins to flex	posterior	Flexor	Absorption (flexors) General (flexors)	iliopsoas, Rectus femoris, iliopsoas, Rectus femoris
Knee	From +5° extends a few degrees, then flexes to about +45°	posterior	Flexors, then Extensors	Absorption (extensors) Small generation (knee flexors) large generation to (Extensors)	Vastus Gastrocnemius Rectus femoris
Ankle	From +5° dorsiflexion, dorsiflexion a few more degrees, then rapidly plantarflexes to -25°	Anterior	plantarflexors	Very small absorption (plantarflexors) large generation (plantarflexors)	Soleus Gastrocnemius Soleus Gastrocnemius

Frontal Plane Analysis				
Th	Motion	GRF	Moment	Power
Hip	From $+3^{\circ}$ smoothly abd to -5°	Medial	Abductor	Generat ⁿ (abduct ⁿ)
Knee	Neutral, then abd to a few degree -3°	Medial	Abductor	Absorption (Abd)
Ankle	—	Inconsistent	—	Gluteus medius Tensor fasciae latae, ligaments

Major Muscle Activity

Transverse Plane Analysis

Th	Motion	Moment	Power
Hip	From -3° , External rotat ⁿ a few more degrees	Internal rotator	Very small absorption (external rotators)
Knee	Remains near Neutral	Internal rotator	Very small absorption (ligaments)
Ankle	—	Inconsistent	—

Gait Initiation and Termination :-

Gait Initiation

It is a stereotyped activity that includes the series or sequence of events that occur from the initiation of movement to the beginning of the gait cycle.

(or)

The phase b/w standing motionless and steady-state locomotion.

→ Gait initiation begins in erect standing posture.

COP - posteriorly

Gait Termination

→ Termination ; transition from walking to standing

→ It has received much less attention than has initiation.

Gait Initiation :-

→ Gastrocnemius and soleus go off.

→ Bilateral concentric activity of Tibialis anterior.

→ COP moves posterior, Briefly towards 1st swing foot.

→ COP moves to stance foot heel → forefoot.

→ During wt transfer on other leg, Abductors (Glu med) will fire up along with tibialis anterior.

→ Some amount of Hip - knee flexion at support limb and COP moves antero-medial towards support limb.

- No change will be seen in pattern of events
- Temporal & Spatial variatⁿ are preserved in elderly (almost).

Gait Termination:

- Anticipation of stopping
- Influences EMG activity.
- Stance limb anterior to whole body COM
achieved by reducing push off (Soleus off & Tibialis ant on)
by activating hip & knee extensors for energy absorption.
- Swing - Vastus lateralis - 1st active and continued to maintain knee.
- Extension along with erector spine.
- Breaking forces by soleus at ankle.

Trunk and Upper Extremities:-

Trunk:

Flexion and Extension-

- low amplitude flexion peak at heel strike.
- low amplitude extension peak at single limb support.
- Unbalancing flexion moment at trunk during initial contact countered by (H₁-S) Hip extensors.

Abduction and Adduction -

- COM and adduction moment is countered by hip abductors to Hip Jt.
- Pelvis Rotates forward in Transverse plane
- Pelvis moves in Sinusoidal curve in Sagittal plane
- Pelvis moves in translates side to side in Frontal plane.

Trunk and Pelvis -

- Both rotates in opposite direction.
- This helps to reduce excessive body movement by Counterbalancing rotation of pelvis.
- At ↑ speed walking, relationship b/w Head and Trunk movement and walking speed is no more linear (less coordinatⁿ).

EMG:

- Internal, External oblique get ^(↑) More activity during initial contact.
- Rectus Abdominus, Lumbar Erector spinae are
(flexion) (extension)
more active during the stance phase (wt acceptⁿ).

Upper Extremities:

- Arm and trunk move in same direction
- Pelvis + legs → same directⁿ
- Trunk + Arms → same directⁿ

→ Arm swing provides counter balancing action to forward swinging of leg by decelerating rotation of the body.

→ Studies also support neuronal co-ordination of arm and leg movement.

Muscle activity: movement → Gravity, muscles.

Forward Arm Swing → Medial rotators, subscapularis, teres major and latissimus dorsi

Backward Arm Swing → Mid & post deltoid also latissimus dorsi, teres major (same part)

→ Supraspinatus, trapezius, deltoid active in both.

→ Medial rotators work eccentrically to control external rotation in forward swing.

Backward swing is produced by concentric activity

Stair Gait:-

→ (Wt acceptance (heel strike) - 14%

stance } Pull up - (14 - 32%) with single limb support

max } Forward continuance (32 - 64%) - mid-stance + toe off

Foot Clearance (64 - 80%)

Foot placement (80 - 100%)

} Swing phase

→ ↑ ROM at hip, knee, ankle

Trunk flexion - (2x) absent

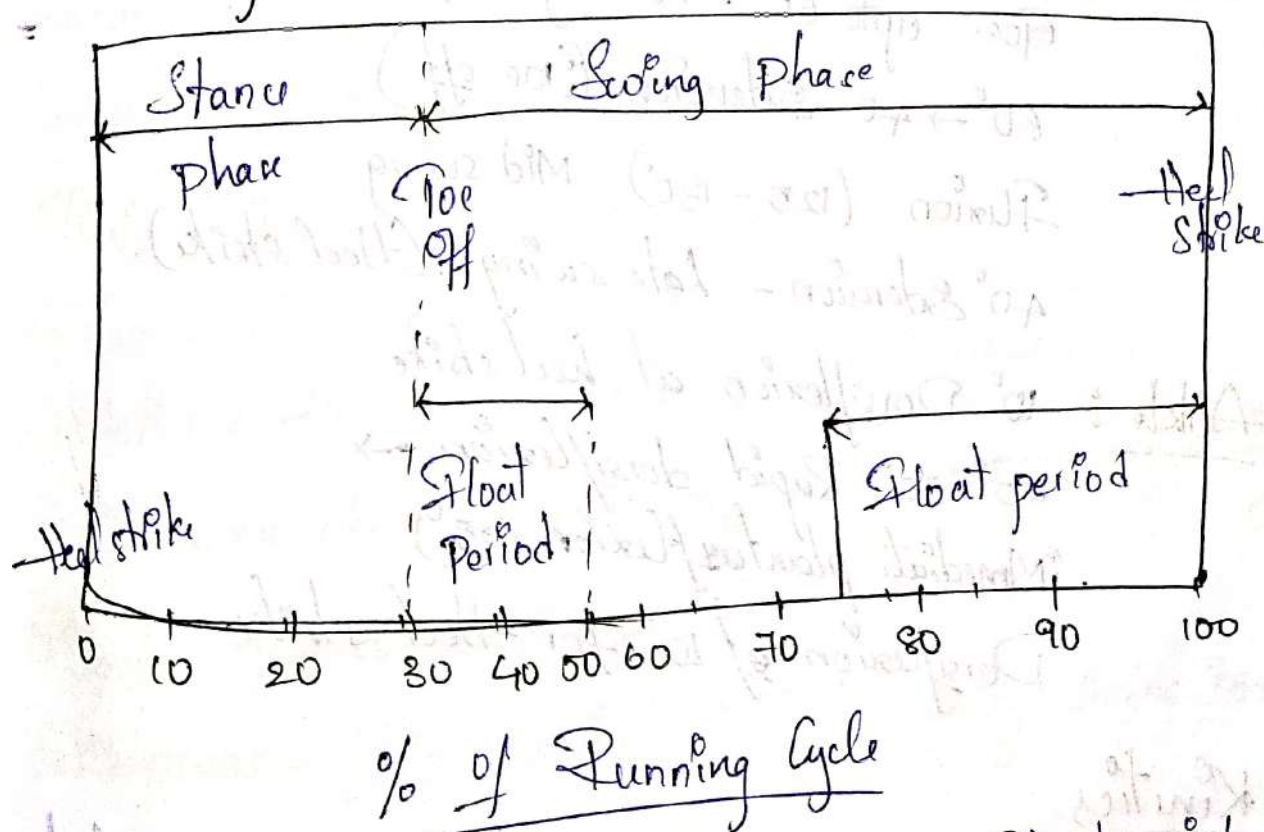
→ Ascending includes large positive/concentric work
Rectus femoris, Vastus lateralis, Gastrocnemius

→ Descending includes large negative/eccentric work
Hip Abductors, Ankle plantar flexors

→ Eccentric activity is similar to normal gait
but (↑ 3x) magnitude.

→ ↑↑ moment can cause mobility issues in OA
and PFPS pts.

Running Gait :-



→ Absence of double support period → Float period
(↑ with speed).

→ GRF at COP ↑ to 200% of body wt, ↑ force on

Patello femoral jt.

→ In running both feet fall in same line of progression.

→ ↑ functional limb varus to improve BOS (15°)

→ ↑ ROM, ↑ Muscle activation, ↓ BOS, More force at

Kinematics

Hip: 45° flexion - Heel strike

20° extension - after toe off

Flexion 60° - late swing

Goes to $45^\circ - 50^\circ$ to prepare for heel strike.

Knee: $20 - 40^\circ$ at heel strike

Goes upto 60° - loading response

$60^\circ \rightarrow 40^\circ$ Extension (toe off)

Flexion ($125 - 130^\circ$) Mid swing

40° Extension - late swing (heel strike).

Ankle: 10° Dorsiflexion at heel strike

$25^\circ - 30^\circ$ Rapid dorsiflexion →

immediate plantar flexion (25°)

Dorsiflexion of 10° for heel strike

Kinetics

Gluteus: Max, Medius and ~~min~~ Tensor Fascia lata

Active in end of swing phase and beginning of stance phase.

Hamstrings: (Medial muscles)

- Concentric (Shortening) - initial and mid stance
- Eccentric (Lengthening) - Controls knee extension (Late swing)
- Also causes Hip extension (Simultaneously).

Quadriceps:

- Eccentric - controls knee flexion (Early stance)
- Concentric - in swing phase to extension
- Knee to prepare for heel strike.

Gastrocnemius:

- Starts at heel strike and continues until 15%.
- Also last 15% of (Swing phase)

Tibialis Anterior:

- Concentric and isometric activity.
- Activity for about 73% of gait cycle,
- ↑ Swing phase in running, hence more time activity

Adductor Magnus - Active for 25%

Iliopsoas - (Hip flexion) Activity during swing (35-60)

Gait initiation

- Gait initiation may be defined as an activity that includes the series or sequence of events that occur from the initiation of movement to the beginning of gait cycle.
- Gait initiation begins in erect standing posture with activation of the *tibialis anterior* & *evastus lateralis* muscles, in conjunction with an inhibition of the *gastroc* muscles, bilateral concentric contraction of the *tibialis anterior* muscles results in a sagittal torque that inclines the body anteriorly from ankles.